

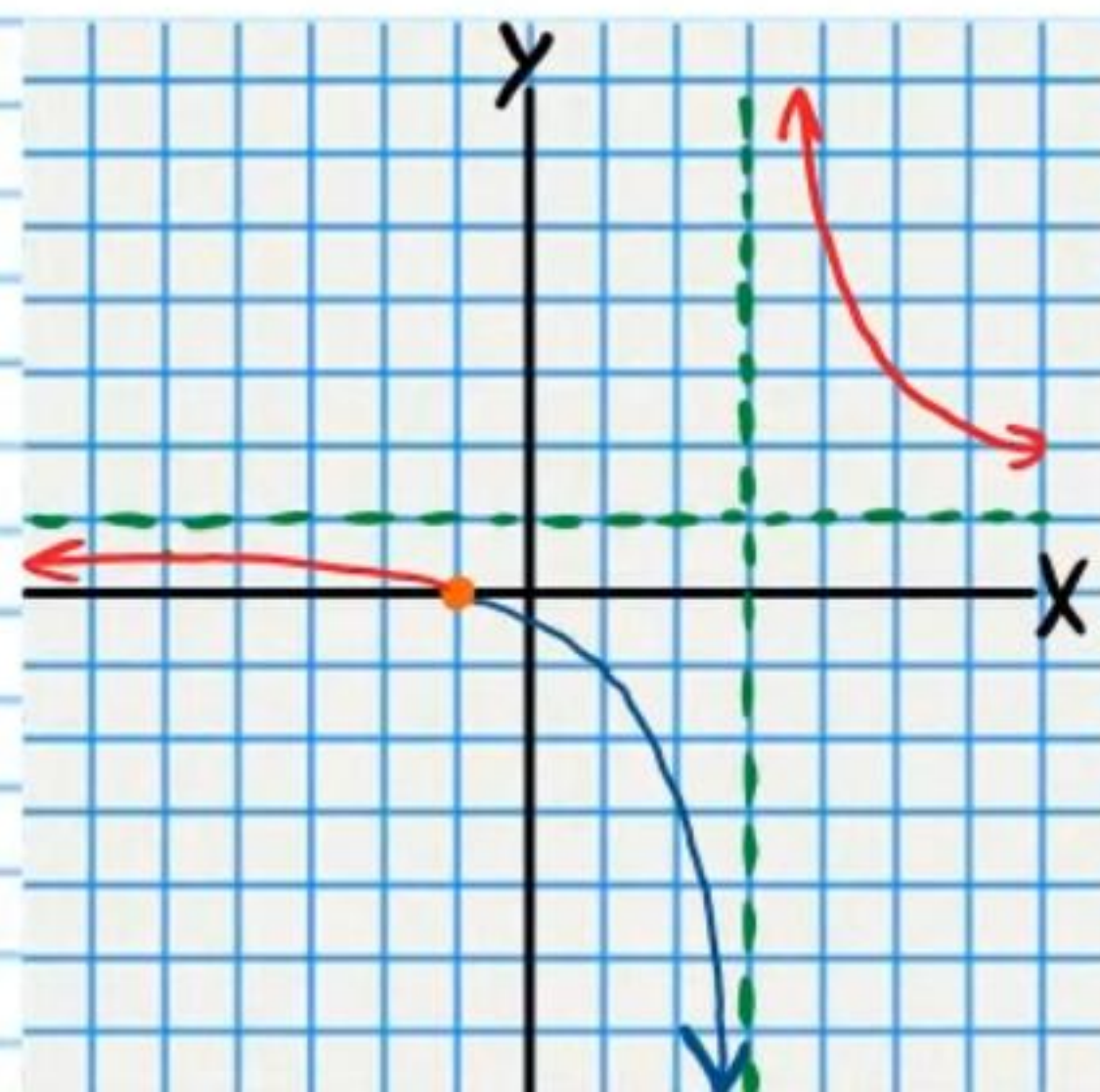
Solving Nonlinear Inequalities (Rational Expressions)

Name
Date
Course

Steps for Solving a Nonlinear Inequality

- I) Move all of the terms to one side of the inequality.
- II) If there is a rational expression (i.e. a fraction with variables in the denominator), combine all of the terms into one fraction.
- III) Find the x-values where the expression is either zero or undefined, and mark these values on a number line.
- IV) Determine the sign of each interval, and use the number line to write the graphical solution of the inequality. Other forms of the solution can be written also. The form most commonly used is interval notation.

Example: $f(x) = \frac{x+1}{x-3}$



Types of Factoring

Greatest Common Factor $\Rightarrow$ Ex: $10x^3 + 5x - 20x^2 = 5x(2x^2 + 1 - 4x)$

Quadratic Expressions $\Rightarrow$ Ex: $x^2 - 5x - 6 = (x-6)(x+1)$

Difference of Two Squares $\Rightarrow$ Ex: $x^2 - 49 = (x+7)(x-7)$

$ax^4 + bx^2 + c$ Form $\Rightarrow$ Ex: $x^4 + 5x^2 + 6 = (x^2+3)(x^2+2)$

Grouping $\Rightarrow$ Ex: $x^3 - 2x^2 + 2x - 4 = x^2(x-2) + 2(x-2) = (x-2)(x^2+2)$

Rational Zeros Theorem $\Rightarrow$ Ex: $x^3 - 2x^2 - 5x + 6$

$P: 6 \Rightarrow 1, 2, 3, 6$ $\frac{P}{q}: \pm 1, \pm 2, \pm 3, \pm 6$ $1 \mid \begin{array}{rrrr} 1 & -2 & -5 & 6 \\ & 1 & -1 & -6 \\ \hline & 1 & -1 & -6 & 0 \end{array}$

$q: 1 \Rightarrow 1$

$x^3 - 2x^2 - 5x + 6 = (x-1)(x^2 - x - 6)$

Sum or Difference of Two Cubes $\Rightarrow$ Ex: $x^3 - 64 = (x-4)(x^2 + 4x + 16)$

Solve. Write the solutions graphically and in interval notation.

$$\textcircled{1} \frac{x^4}{x-1} + \frac{4x^3}{x-1} \geq -\frac{4x^2}{x-1}$$

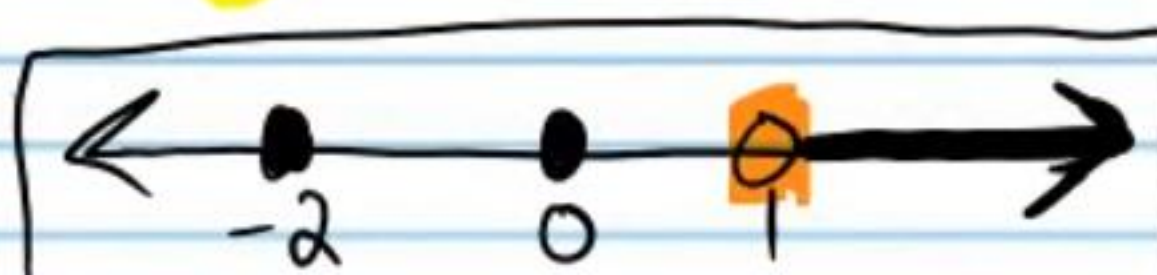
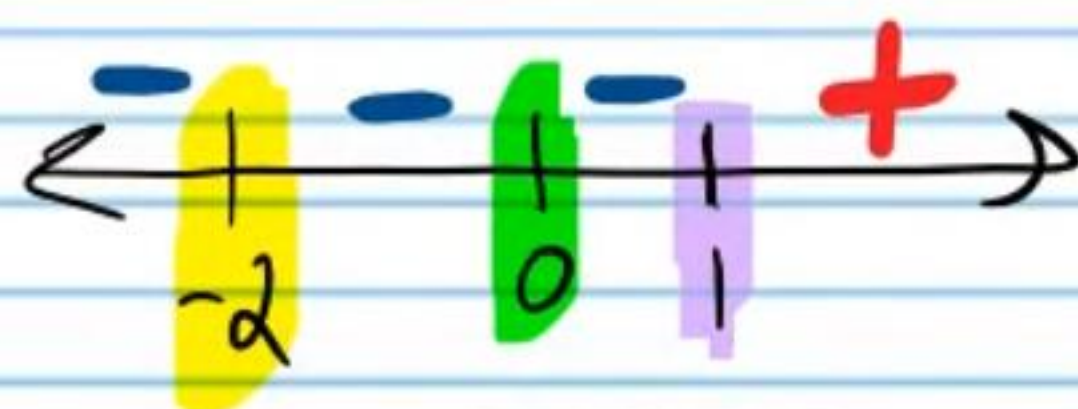
$$\frac{x^4}{x-1} + \frac{4x^3}{x-1} + \frac{4x^2}{x-1} \geq 0$$

$$\frac{x^4 + 4x^3 + 4x^2}{x-1} \geq 0$$

$$\frac{x^2(x^2 + 4x + 4)}{x-1} \geq 0$$

$$\frac{x^2(x+2)(x+2)}{x-1} \geq 0$$

$$\frac{x^2(x+2)^2}{x-1} \geq 0$$



$$[-2, -2] \cup [0, 0] \cup (1, \infty)$$

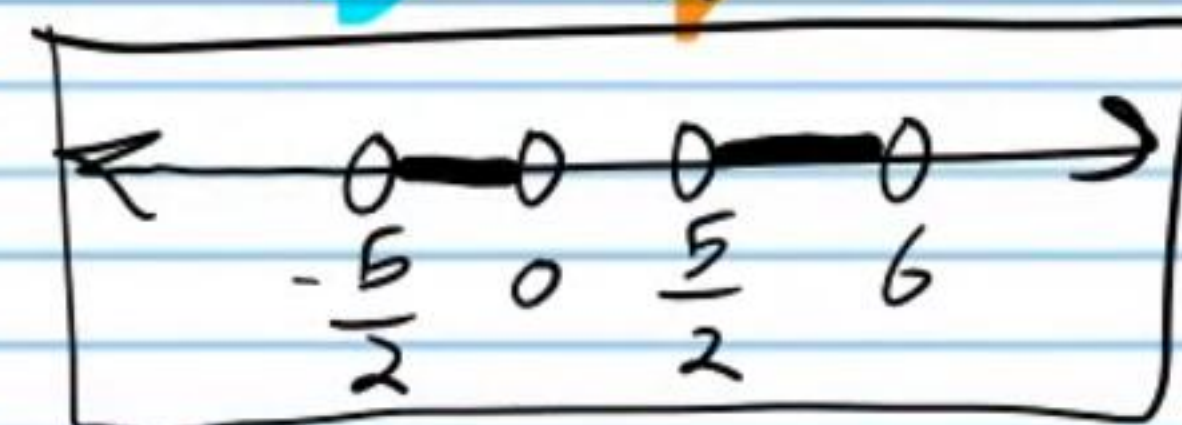
$$\textcircled{2} \frac{3x-6}{20x^3-125x} < \frac{5x+1}{20x^3-125x} - \frac{3x+1}{20x^3-125x}$$

$$0 < \frac{5x+1}{20x^3-125x} - \frac{3x+1}{20x^3-125x} - \frac{3x-6}{20x^3-125x}$$

$$0 < \frac{-x+6}{20x^3-125x}$$

$$0 < \frac{-x+6}{5x(4x^2-25)}$$

$$0 < \frac{-x+6}{5x(2x+5)(2x-5)}$$



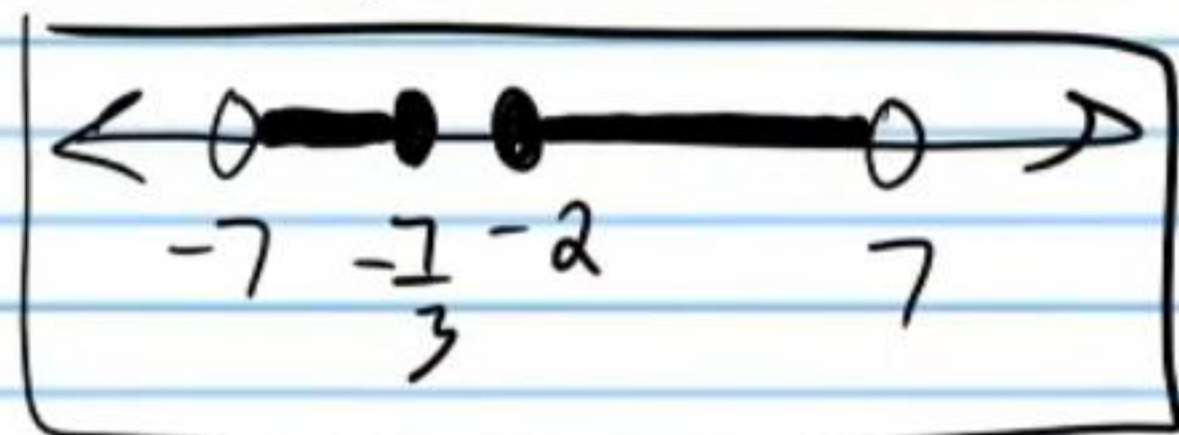
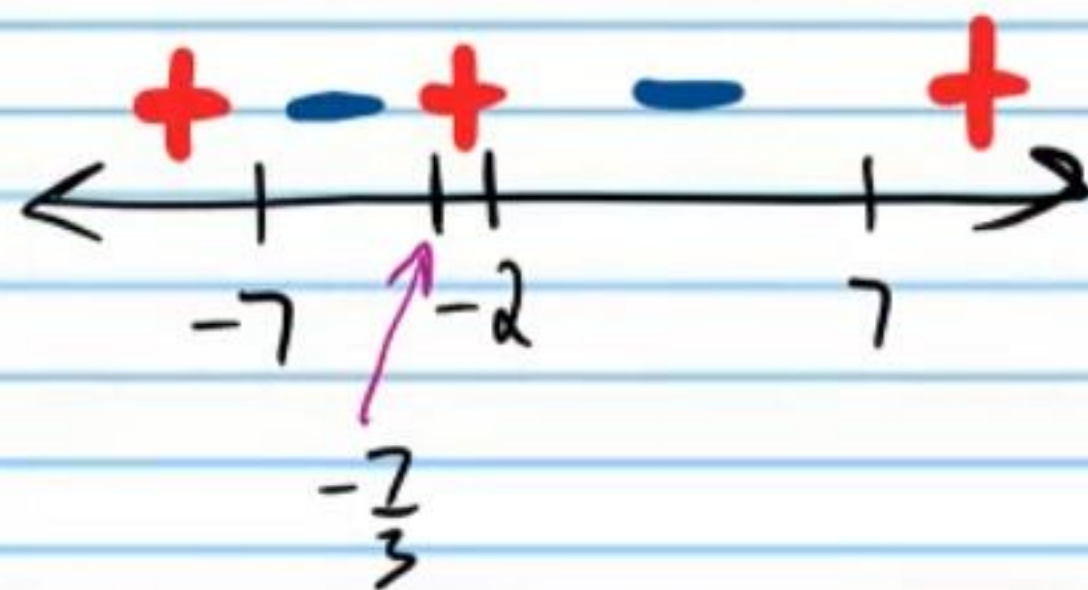
$$(-\frac{5}{2}, 0) \cup (\frac{5}{2}, 6)$$

$$\textcircled{3} \frac{11x^2+12x}{x^2-49} + \frac{-8x^2+x}{x^2-49} \leq -\frac{14}{x^2-49}$$

$$\frac{11x^2+12x}{x^2-49} + \frac{-8x^2+x}{x^2-49} + \frac{14}{x^2-49} \leq 0$$

$$\frac{3x^2+13x+14}{x^2-49} \leq 0$$

$$\frac{(3x+7)(x+2)}{(x+7)(x-7)} \leq 0$$



$$\boxed{(-7, -\frac{2}{3}] \cup [-2, 7)}$$

$$\bullet \textcircled{4} \frac{4x^4}{x+5} > \frac{100x^2}{x+5} + \frac{300x^2}{x+5}$$

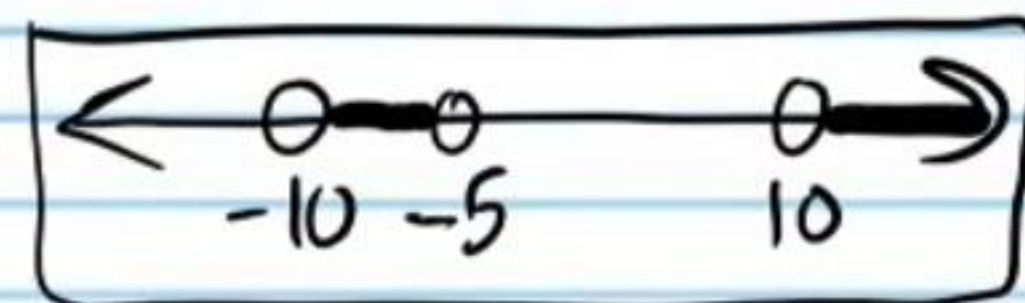
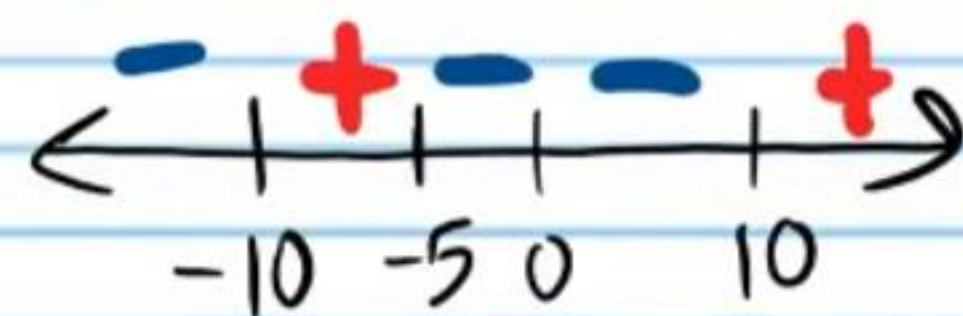
$$\frac{4x^4}{x+5} > \frac{400x^2}{x+5}$$

$$\frac{4x^4}{x+5} - \frac{400x^2}{x+5} > 0$$

$$\frac{4x^4-400x^2}{x+5} > 0$$

$$\frac{4x^2(x^2-100)}{x+5} > 0$$

$$\frac{4x^2(x+10)(x-10)}{x+5} > 0$$



$$\boxed{(-10, -5) \cup (10, \infty)}$$

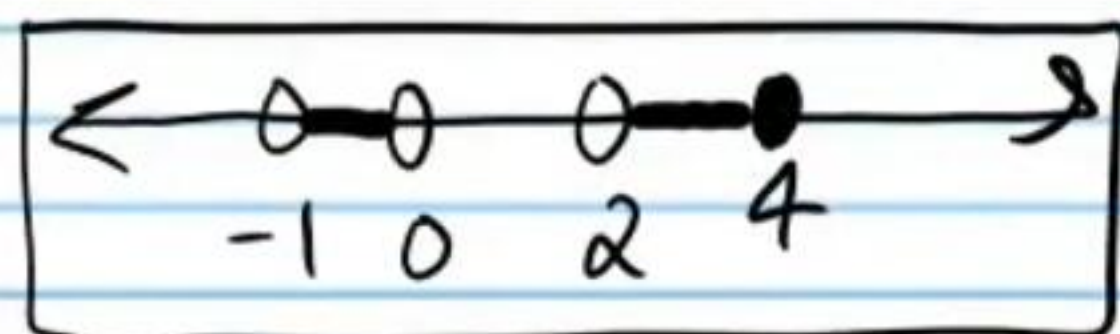
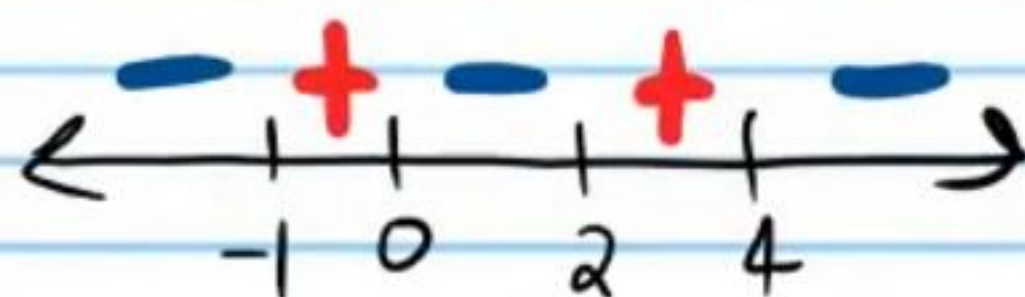
$$\textcircled{5} \frac{x}{-x^3+x^2+2x} \leq \frac{3x+5}{-x^3+x^2+2x} - \frac{x+9}{-x^3+x^2+2x}$$

$$0 \leq \frac{3x+5}{-x^3+x^2+2x} - \frac{x+9}{-x^3+x^2+2x} - \frac{x}{-x^3+x^2+2x}$$

$$0 \leq \frac{x-4}{-x^3+x^2+2x}$$

$$0 \leq \frac{x-4}{-x(x^2-x-2)}$$

$$0 \leq \frac{x-4}{-x(x-2)(x+1)}$$



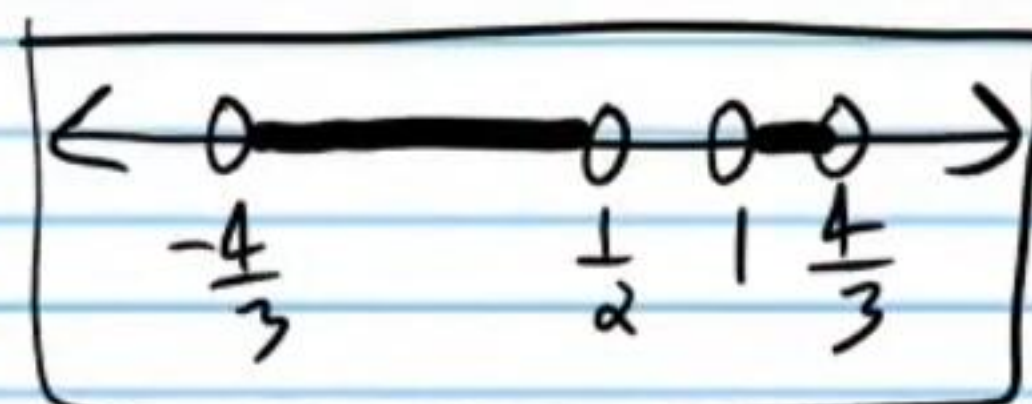
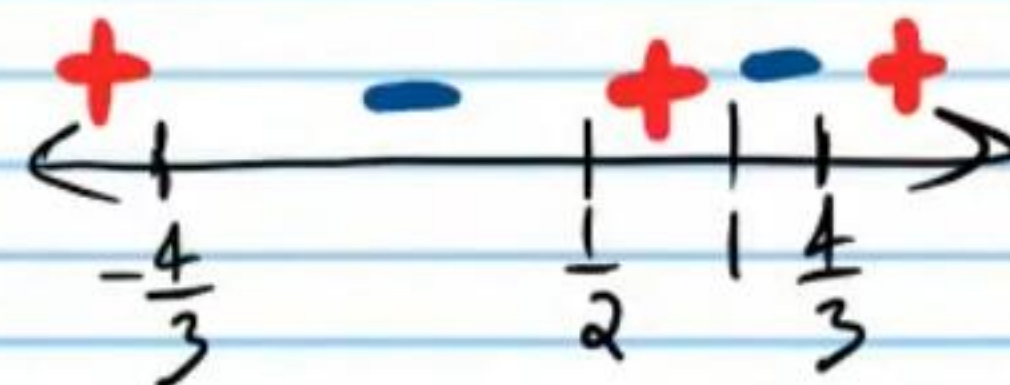
$$(-1, 0) \cup (2, 4]$$

$$\textcircled{6} \frac{2x^2}{9x^2-16} + \frac{10x+1}{9x^2-16} < \frac{13x}{9x^2-16}$$

$$\frac{2x^2}{9x^2-16} + \frac{10x+1}{9x^2-16} - \frac{13x}{9x^2-16} < 0$$

$$\frac{2x^2-3x+1}{9x^2-16} < 0$$

$$\frac{(2x-1)(x-1)}{(3x+4)(3x-4)} < 0$$

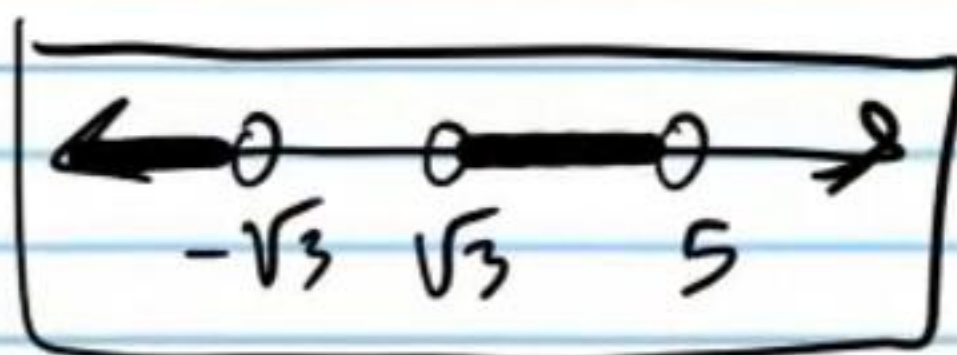
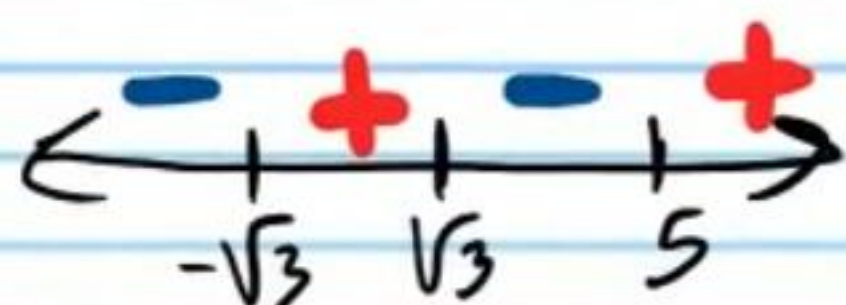


$$(-\frac{4}{3}, \frac{1}{2}) \cup (1, \frac{4}{3})$$

$$\textcircled{7} \frac{x^4 + 2x^2 - 15}{x^3 - 5x^2 + 2x - 10} < 0$$

$$\frac{(x^2 - 3)(x^2 + 5)}{x^2(x - 5) + 2(x - 5)} < 0$$

$$\frac{(x + \sqrt{3})(x - \sqrt{3})(x^2 + 5)}{(x - 5)(x^2 + 2)} < 0$$



$$(-\infty, -\sqrt{3}) \cup (\sqrt{3}, 5)$$

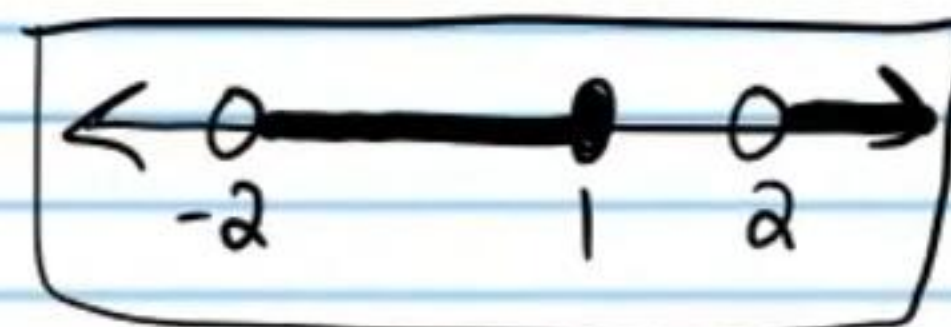
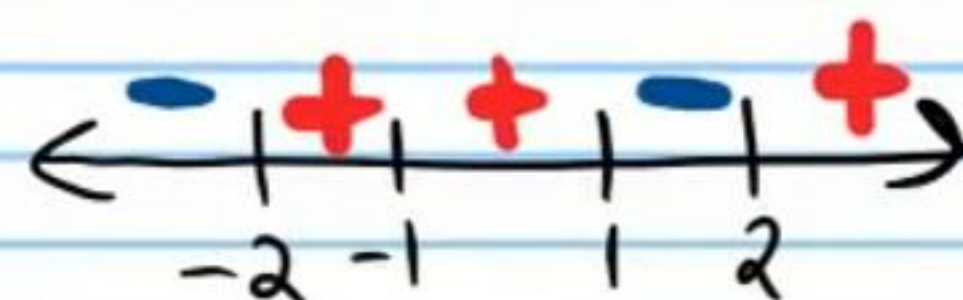
$$\textcircled{8} \frac{x^3 + x^2 - x - 1}{x^4 - 2x^2 - 8} \geq 0$$

$$\frac{x^2(x+1) - (x+1)}{(x^2 - 4)(x^2 + 2)} \geq 0$$

$$\frac{(x+1)(x^2 - 1)}{(x+2)(x-2)(x^2 + 2)} \geq 0$$

$$\frac{(x+1)(x+1)(x-1)}{(x+2)(x-2)(x^2 + 2)} \geq 0$$

$$\frac{(x+1)^2(x-1)}{(x+2)(x-2)(x^2 + 2)} \geq 0$$



$$(-2, 1] \cup (2, \infty)$$

Factoring the Sum or Difference of Two Cubes

Factoring the Sum or Difference of Two Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

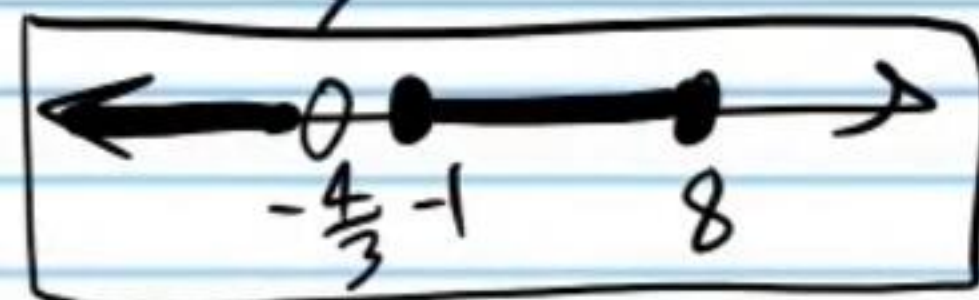
$$\textcircled{9} \frac{x^2 - 7x - 8}{27x^3 + 64} \leq 0$$

$$\frac{(x-8)(x+1)}{(3x)^3 + 4^3} \leq 0$$

$$\frac{(x-8)(x+1)}{(3x+4)((3x)^2 - (3x)(4) + 4^2)} \leq 0$$

$$\frac{(x-8)(x+1)}{(3x+4)(9x^2 - 12x + 16)} \leq 0$$

$$D = b^2 - 4ac = (-12)^2 - 4(9)(16) = 144 - 576 = -432 \quad \times$$



$$(-\infty, -\frac{4}{3}) \cup [-1, 8]$$

$$\textcircled{10} \frac{27 - x^3}{4x^2 - 25} \leq 0$$

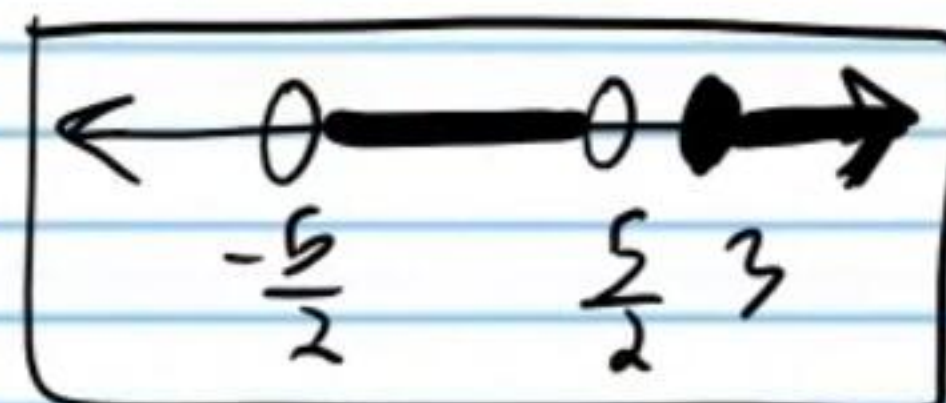
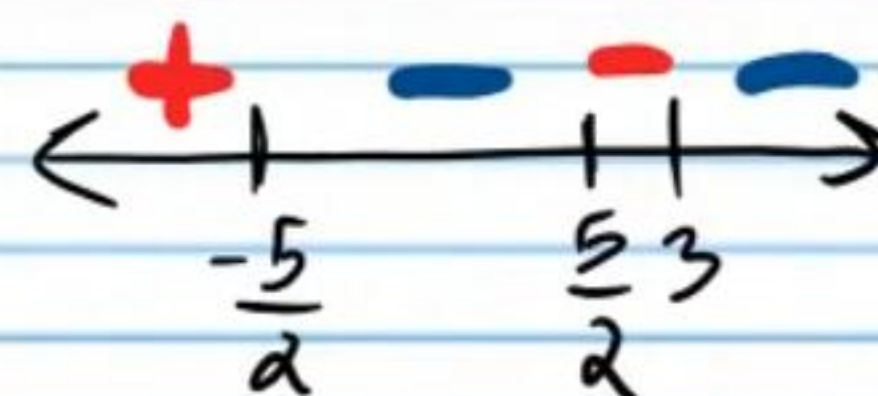
$$\frac{3^3 - x^3}{(2x+5)(2x-5)} \leq 0$$

$$\frac{(3-x)(3^2 - 3x + x^2)}{(2x+5)(2x-5)} \leq 0$$

$$\frac{(3-x)(9 - 3x + x^2)}{(2x+5)(2x-5)} \leq 0$$

$$\frac{(3-x)(x^2 - 3x + 9)}{(2x+5)(2x-5)} \leq 0$$

$$D = b^2 - 4ac = (-3)^2 - 4(1)(9) = 9 - 36 = -27 \quad \times$$



$$(-\frac{5}{2}, \frac{5}{2}) \cup [3, \infty)$$

$$\textcircled{11} \frac{x+2}{x-4} \leq 3$$

$$\frac{x+2}{x-4} - 3 \leq 0$$

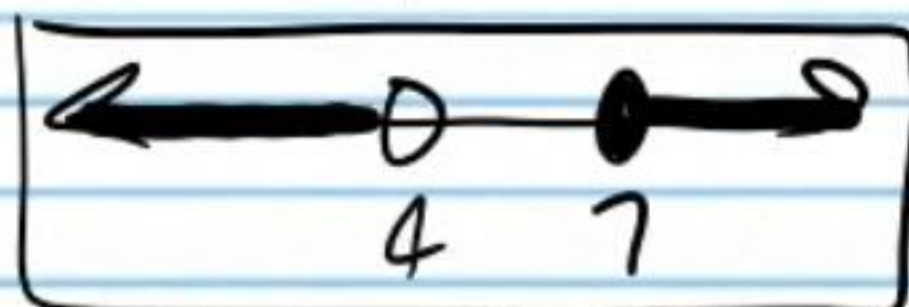
$$\frac{x+2}{x-4} - \frac{3}{1} \leq 0$$

$$\frac{x+2}{x-4} - \frac{3(x-4)}{1(x-4)} \leq 0$$

$$\frac{x+2}{x-4} - \frac{3x-12}{x-4} \leq 0$$

$$\frac{-2x+14}{x-4} \leq 0$$

$$\frac{-2(x-7)}{x-4} \leq 0$$



$$(-\infty, 4) \cup [7, \infty)$$

$$\textcircled{12} \frac{3}{x} < 2+x$$

$$0 < 2+x - \frac{3}{x}$$

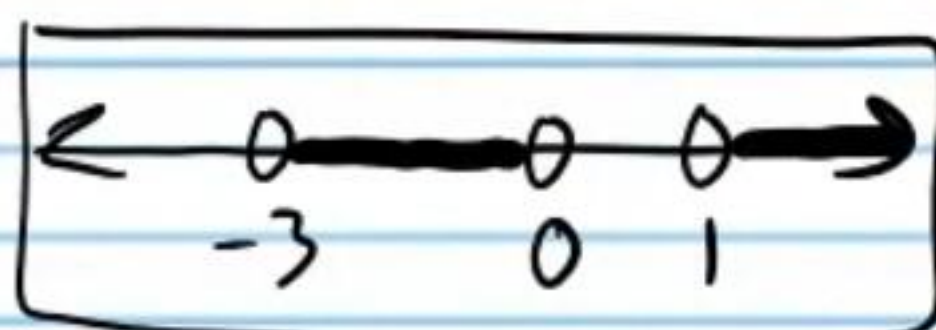
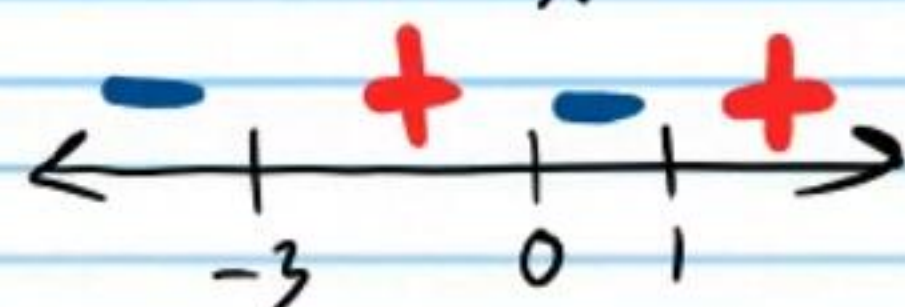
$$0 < \frac{2}{1} + \frac{x}{1} - \frac{3}{x}$$

$$0 < \frac{2 \cdot x}{1 \cdot x} + \frac{x \cdot x}{1 \cdot x} - \frac{3}{x}$$

$$0 < \frac{2x}{x} + \frac{x^2}{x} - \frac{3}{x}$$

$$0 < \frac{x^2+2x-3}{x}$$

$$0 < \frac{(x-1)(x+3)}{x}$$



$$(-3, 0) \cup (1, \infty)$$

$$\textcircled{13} \frac{x-1}{2x+3} > \frac{1}{x+5}$$

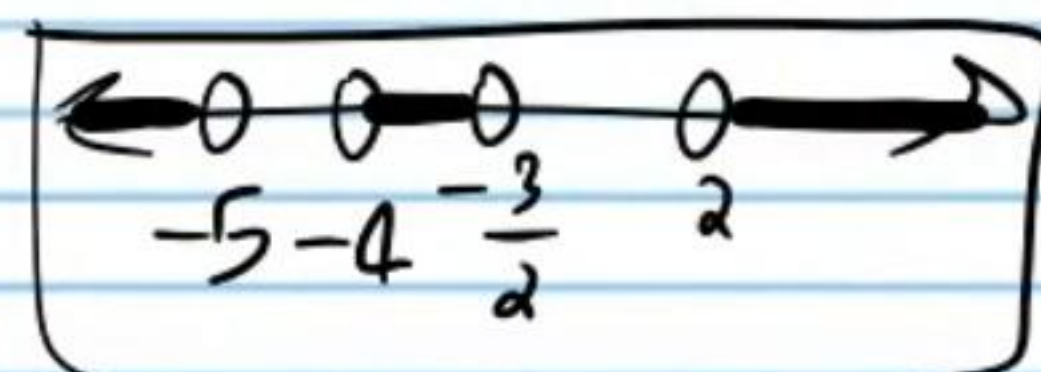
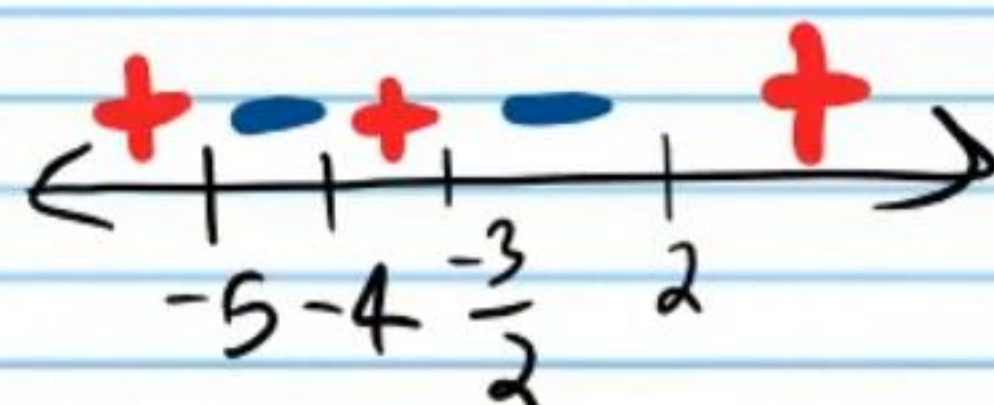
$$\frac{x-1}{2x+3} - \frac{1}{x+5} > 0$$

$$\frac{(x-1)(x+5)}{(2x+3)(x+5)} - \frac{1(2x+3)}{(x+5)(2x+3)} > 0$$

$$\frac{x^2+4x-5}{(2x+3)(x+5)} - \frac{2x+3}{(x+5)(2x+3)} > 0$$

$$\frac{x^2+2x-8}{(2x+3)(x+5)} > 0$$

$$\frac{(x+4)(x-2)}{(2x+3)(x+5)} > 0$$



$$(-\infty, -5) \cup (-4, -\frac{3}{2}) \cup (2, \infty)$$

$$\textcircled{14} -1 > \frac{x+5}{x+1}$$

$$0 > \frac{x+5}{x+1} + 1$$

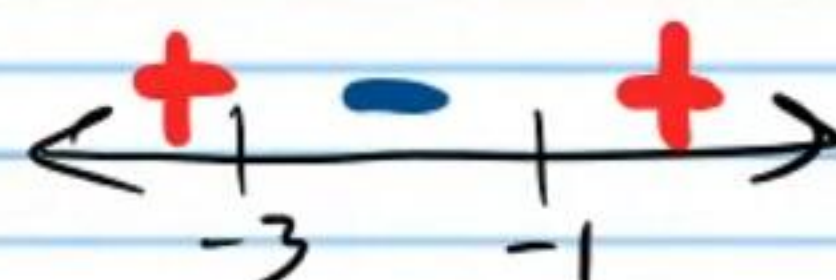
$$0 > \frac{x+5}{x+1} + \frac{1}{1}$$

$$0 > \frac{x+5}{x+1} + \frac{1(x+1)}{1(x+1)}$$

$$0 > \frac{x+5}{x+1} + \frac{x+1}{x+1}$$

$$0 > \frac{2x+6}{x+1}$$

$$0 > \frac{2(x+3)}{x+1}$$



$$(-3, -1)$$

$$\textcircled{15} \quad x - \frac{28}{x} \geq 3$$

$$x - \frac{28}{x} - 3 \geq 0$$

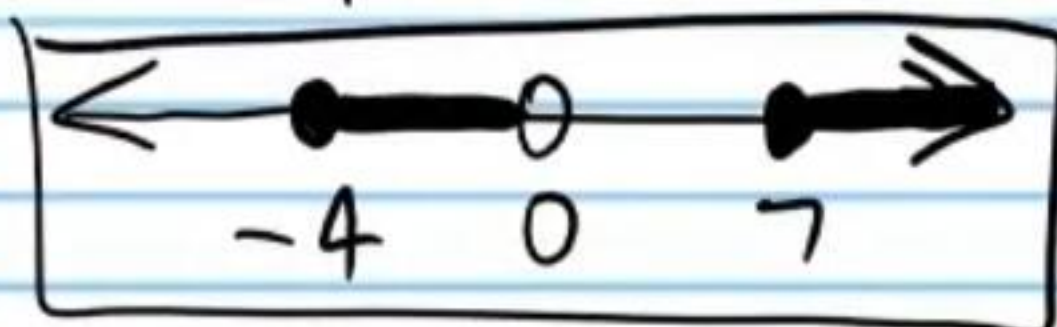
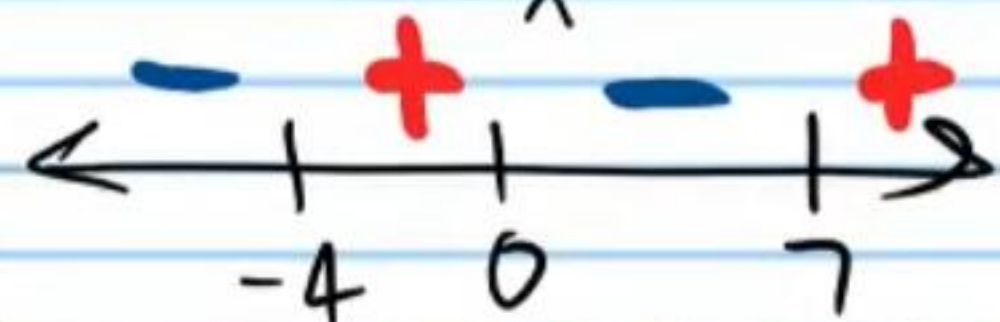
$$\frac{x}{1} - \frac{28}{x} - \frac{3}{1} \geq 0$$

$$\frac{x \cdot x}{1 \cdot x} - \frac{28}{x} - \frac{3 \cdot x}{1 \cdot x} \geq 0$$

$$\frac{x^2}{x} - \frac{28}{x} - \frac{3x}{x} \geq 0$$

$$\frac{x^2 - 3x - 28}{x} \geq 0$$

$$\frac{(x-7)(x+4)}{x} \geq 0$$



$$[-4, 0) \cup [7, \infty)$$

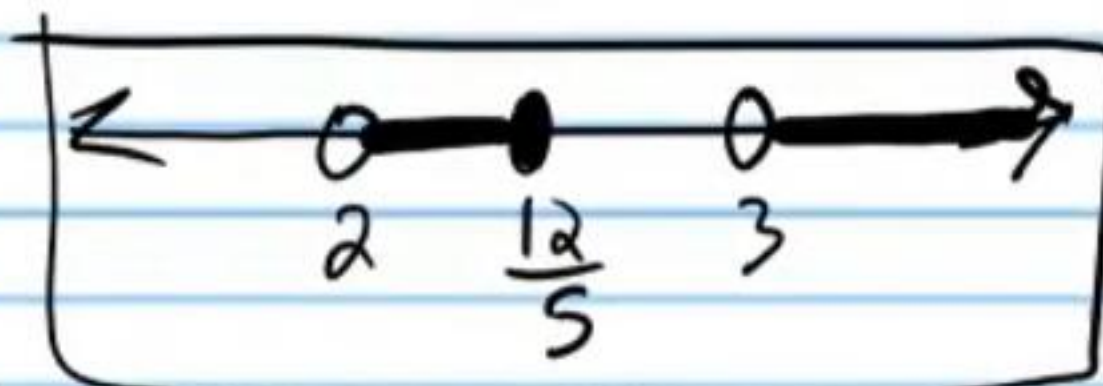
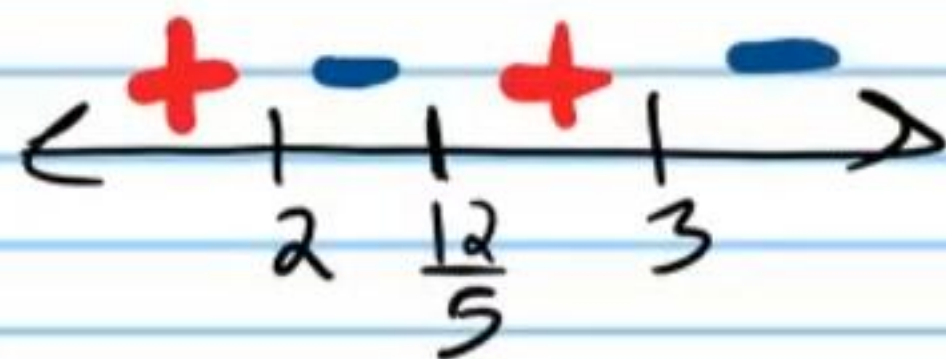
$$\textcircled{16} \quad \frac{x-6}{x-3} \leq \frac{x}{x-2}$$

$$\frac{x-6}{x-3} - \frac{x}{x-2} \leq 0$$

$$\frac{(x-6)(x-2)}{(x-3)(x-2)} - \frac{x(x-3)}{(x-2)(x-3)} \leq 0$$

$$\frac{x^2 - 8x + 12}{(x-3)(x-2)} - \frac{x^2 - 3x}{(x-2)(x-3)} \leq 0$$

$$\frac{-5x + 12}{(x-3)(x-2)} \leq 0$$



$$(2, \frac{12}{5}] \cup (3, \infty)$$